

Disproof of TLMC Conjecture 00000000485

The proportion of shapes with $|\chi^\lambda(c_n)| > 1$ does not tend to 1

TLMC falsification submission

October 2, 2026

1 The conjecture

For a partition $\lambda \vdash n$, let χ^λ be the irreducible character of the symmetric group S_n attached to λ , and let $c_n \in S_n$ be a cycle of length n .

Conjecture 1 (00000000485). *The proportion of shapes $\lambda \vdash n$ with $|\chi^\lambda(c_n)| > 1$ tends to 1 as $n \rightarrow \infty$.*

2 The attack: n-cycle characters are always in $\{0, \pm 1\}$

Recall the Murnaghan–Nakayama rule: $\chi^\lambda(\mu)$ is a signed sum over all rim hooks (border strips) of size μ_1 that can be removed from λ , continued recursively. A rim hook is a connected skew shape containing no 2×2 block, and each contributes the sign $(-1)^{h-1}$, where h is its number of rows.

Take $\mu = (n)$, i.e. the conjugacy class of long cycles c_n . Removing a rim hook of size n from a diagram with exactly n boxes means removing the *entire diagram*. This is possible if and only if λ itself is a border strip. For a straight (non-skew) shape this means precisely that λ contains no 2×2 block, i.e. $\lambda_2 \leq 1$, i.e. λ is a *hook*

$$\lambda = (n - k, 1^k) \quad (0 \leq k \leq n - 1),$$

in which case the whole diagram has $h = k + 1$ rows and

$$\chi^\lambda(c_n) = (-1)^{h-1} = (-1)^k.$$

If λ is not a hook, no removal is possible and $\chi^\lambda(c_n) = 0$.

Theorem 1 (classical). *For every $n \geq 1$ and every $\lambda \vdash n$,*

$$\chi^\lambda(c_n) \in \{-1, 0, +1\}.$$

Explicitly, $\chi^\lambda(c_n) = (-1)^k$ if $\lambda = (n - k, 1^k)$ is a hook, and 0 otherwise.

Proposition 1. *Conjecture 00000000485 is false. In fact*

$$\#\{\lambda \vdash n : |\chi^\lambda(c_n)| > 1\} = 0 \quad \text{for every } n,$$

so the proportion in question is the constant function 0, and $0 \not\rightarrow 1$.

Proof. Immediate from the theorem: no shape ever satisfies $|\chi^\lambda(c_n)| > 1$. □

The conjecture is therefore maximally false: the quantity it asserts tends to 1 is identically 0. There are no boundary cases — the bound $|\chi^\lambda(c_n)| \leq 1$ is unconditional for all n and all $\lambda \vdash n$.

3 Explicit counts

For small n , the number $N(n)$ of shapes with $|\chi^\lambda(c_n)| > 1$, out of $p(n)$ shapes:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	15
$p(n)$	1	2	3	5	7	11	15	22	30	42	56	77	101	176
$N(n)$	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 1: The proportion with $|\chi^\lambda(c_n)| > 1$ is 0 for every n .

For example, at $n = 10$: the hooks $(10), (9, 1), \dots, (1^{10})$ give $\chi = 1, -1, 1, -1, \dots$ (alternating with k), and the remaining 32 shapes all give 0.

4 Verification

Independent recomputation. A standalone Python program (`reproduce.py`) recomputes all n -cycle characters from scratch via recursive rim-hook removal, and validates the implementation against three independent facts: column orthogonality $\sum_\lambda \chi^\lambda(\mu)^2 = z_\mu$ for every $\mu \vdash n$; the hook-length formula $f^\lambda = \chi^\lambda(1^n)$; and the standard-representation identity $\chi^{(n-1,1)}(\mu) = \#\text{Fix}(\mu) - 1$. All checks pass for $n \leq 15$; the n -cycle column contains only $0, \pm 1$, and $N(n) = 0$ throughout.

Lean 4 machine check. The `lean4/` package contains a computable model of the Murnaghan–Nakayama value at the long cycle (exact partition enumeration, border-strip test, alternating sign) and kernel-checked theorems, with *zero axioms* and *zero sorry*: the enumerations have the correct cardinalities (`(parts 10).length = 42`, `(parts 12).length = 77`) and every enumerated part sums to n ; the number of shapes with $|\chi| > 1$ equals 0 for $n = 3, \dots, 12$; and `#print axioms` reports that no listed theorem depends on any axiom.

5 Conclusion

Conjecture 00000000485 is FALSE: $\Pr_{\lambda \vdash n} [|\chi^\lambda(c_n)| > 1] \equiv 0 \not\equiv 1$.